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MULTIMEDIA SUPPORT AND MAINTENANCE WEEK 12

## 1. Case Study: Cost-Benefit Analysis of Preventive vs. Reactive Maintenance (Digital Signage Network)

**Scenario Overview**

Consider a retail company operating a **digital signage network of 50 screens** across multiple stores. These screens display promotions and advertisements critical to daily sales. The company must choose between **reactive maintenance (RM)**—repairing failures when they occur—and **preventive maintenance (PM)**—scheduled servicing and monitoring.

# Cost Analysis

## 1. Reactive Maintenance (RM) Costs

Reactive maintenance appears cheaper initially but results in unpredictable and often higher long-term costs.

- **Average repair cost per failure:** $500
- **Failures per year (per screen):** 2
- **Total annual failures:** 50 × 2 = 100
- **Repair cost:** 100 × $500 = **$50,000**

Additional hidden costs:

- **Emergency service premiums:** Repairs can cost **3–5× more than preventive work** (Stratus Unlimited)
- **Downtime losses:** Assume each failure causes 4 hours downtime, costing $50/hour in lost sales:
  - 100 failures × 4 × $50 = **$20,000**
- **Shortened hardware lifespan:** Screens replaced every 4 years instead of 6 years
  - Replacement cost impact: approx. **$15,000/year extra**

☞ **Total RM Annual Cost ≈ $85,000**

### 2. Preventive Maintenance (PM) Costs

Preventive maintenance includes scheduled inspections, cleaning, software updates, and minor repairs.

- **PM contract per screen/year:** $300
- **Total PM cost:** 50 × $300 = **$15,000**

Reduced failure rate:

- Failures drop by ~60% (industry average assumption)
- New failures: 40 per year
- Repair cost: 40 × $500 = **$20,000**

Reduced downtime:

- 40 × 4 × $50 = **$8,000**

Extended hardware lifespan:

- Saves approx. **$10,000/year**
- **Total PM Annual Cost ≈ $33,000**

## Cost Savings Calculation

- **Reactive Maintenance:** $85,000/year
- **Preventive Maintenance:** $33,000/year

### Annual Savings with PM:

**$85,000 – $33,000 = $52,000 saved per year**

This aligns with industry findings that preventive maintenance can reduce costs by **12–18% or more and avoid major future expenses** (Makula).

## Key Benefits of Preventive Maintenance

### 1. Reduced Downtime

PM minimizes unexpected failures, ensuring continuous operation and avoiding lost revenue. (Poppulo)

### 2. Lower Repair Costs

Routine servicing prevents major breakdowns, avoiding expensive emergency repairs.

### 3. Extended Equipment Life

Regular maintenance increases hardware lifespan, delaying capital replacement costs. (USSI Global)

### 4. Improved Reliability & Brand Image

Functional displays maintain customer trust and consistent communication.

# Conclusion

While reactive maintenance may seem cost-effective initially, it results in **higher long-term costs due to downtime, emergency repairs, and faster equipment wear**. Preventive maintenance, although requiring upfront investment, delivers **significant cost savings, improved reliability, and better asset utilization**.

in this case, adopting PM results in **over $50,000 annual savings**, making it the financially and operationally superior strategy for managing a multimedia (digital signage) setup.

# 2. Home Multimedia Preventive Maintenance Kit & Guide

## 1. Essential Maintenance Kit (Tools & Supplies)

A basic multimedia preventive maintenance (PM) kit should include:

### Cleaning Tools

- **Compressed air can (anti-dust, moisture-free)** – for blowing dust out of vents

- **Microfiber cloths** – for screens and surfaces (non-scratch)
- **Soft brush (anti-static)** – for loosening dirt in ports and keyboards

### Cleaning Solutions

- **Isopropyl alcohol (70%–90%)** – safe for electronics (used sparingly)
- **Screen-safe cleaning spray** – for TVs and monitors

### Hand Tools

- **Precision screwdriver set (Phillips & flathead)** – for minor external adjustments
- **Small flashlight** – to inspect ports and cables

### Cable Management

- **Cable ties or Velcro straps** – to organize cables
- **Label stickers or tags** – to identify cables

### Protection & Safety

- **Surge protector or power strip** – protects devices from power spikes
- **Anti-static wrist strap (optional)** – prevents static damage
- **Soft mat or cloth** – to place devices during cleaning

# 2. Simple One-Page Maintenance Guide (Non-Technical Users)

## Safety First

- Always **turn off and unplug** your computer and TV before cleaning
- Never spray liquid directly onto devices
- Avoid opening devices (internal repairs require a technician)

## Step 1: Clean the Outside

- Use a **dry microfiber cloth** to wipe dust from the screen and body
- For smudges, lightly dampen the cloth with **screen cleaner**

- Use **compressed air** to blow dust from vents (short bursts only)

Tip: Hold the air can upright to avoid liquid spray

## Step 2: Clean Ports & Keyboard (Computer)

- Gently blow air into USB ports and keyboard gaps
- Use a **soft brush** to remove stuck dirt

Do not insert sharp objects into ports

## Step 3: Check Cables

- Ensure all cables are **firmly connected**
- Look for **damaged or frayed wires**
- Organize cables using **Velcro ties** to prevent tangling

## Step 4: Improve Airflow

- Keep your computer and TV in a **well-ventilated area**
- Leave at least **5–10 cm space** around vents
- Avoid placing devices on soft surfaces like beds or carpets

Step 5: Power Protection

- Plug devices into a **surge protector** instead of directly into the wall
- Turn off devices during thunderstorms or long periods of non-use

## Step 6: Basic System Care (Computer)

- Delete unnecessary files to free space
- Keep your system updated (Windows updates, antivirus)
- Restart your computer at least **once a week**

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# Step 7: Routine Schedule

- **Weekly:** Wipe surfaces, check cables
- **Monthly:** Clean vents and ports with compressed air
- **Every 3–6 months:** Deep external clean and setup check

- Regular cleaning prevents overheating and improves performance
- Never use water or household cleaners on electronics
- If you notice unusual noise, overheating, or screen issues—seek professional help

## Conclusion

With a simple kit and routine, you can **extend the life of your computer and TV, reduce breakdowns, and maintain smooth performance**—all without needing technical expertise.